# Scales and multimodal flux distributions in stationary metabolic network models via thermodynamics

Daniele De Martino
*Institute of Science and Technology Austria, Am Campus 1, A-3400 Klosterneuburg, Austria*
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In this work it is shown that scale-free tails in metabolic flux distributions inferred in stationary models are an artifact due to reactions involved in thermodynamically unfeasible cycles, unbounded by physical constraints and in principle able to perform work without expenditure of free energy. After implementing thermodynamic constraints by removing such loops, metabolic flux distributions scale meaningfully with the physical limiting factors, acquiring in turn a richer multimodal structure potentially leading to symmetry breaking while optimizing for objective functions.

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## I. INTRODUCTION

Since its modern foundation, physics has given powerful tools to assess scales in biological systems [1]. More recent developments in statistical mechanics [2] have provided new insights on scales' emergence with subsequent applications to biology, an interesting debated case being scale-free dynamics in neural networks as connected to critical behavior [3–6]. A biological process with deep physical roots ideally at the heart of emergence of scales in living systems is metabolism, i.e., the fundamental set of reactions at the core of cellular energetics. The energetic constraints of metabolism have been considered to play a role in many important issues, from eucaryotes' evolution to aging, all the way down to the origin of life itself [7]. Nowadays it is possible to keep track for a given organism of the entire set of enzymes devoted to metabolic functions to reconstruct the whole chemical reaction network at the genome-scale level. The analysis of such complex systems is difficult, and simplifying assumptions are made into a framework known as flux balance analysis [8] with wide applications, e.g., in metabolic engineering [9]. Based on inference on these stationary models, it has been argued that metabolic fluxes are globally organized in a *scale-free* manner around a backbone of reactions carrying high-intensity fluxes [10]. While we acknowledge the existence and importance of central core pathways in metabolism, it is believed that such a core has *scales* simply given by the physical limiting factors, being extrinsic (resources availability [11]) or intrinsic (maximum ribosome elongation rate [12]). Further, it is generally known in the field of flux balance analysis that high-intensity fluxes retrieved in models can be a spurious effect due to an erroneous handling of thermodynamics constraints. In essence, as will be detailed in the following, models of metabolic networks come with arbitrary bounds on the fluxes, upon which the solution space could depend without a proper implementation of thermodynamics and thus hampering in turn any assessment on flux scales.

The impact of these constraints on the space of feasible states [13] is thus important and has attracted attention in recent years [14] to the deeper clarification of its theoretical basis from a mathematical [15] and physical point of view [16].

In this work it will be shown that once thermodynamic constraints are implemented in a minimal way, i.e., by requiring the existence of a free energy gradient or, equivalently, the unfeasibility of closed reaction loops, the long tail in a flux intensity distribution is truncated and the whole space scales meaningfully with the physical limiting factors, e.g., the availability of resources.

On the other hand it is pointed out that the space is not convex anymore, thus acquiring a rich multimodal structure possibly leading to symmetry breaking upon optimizing for objective functions. After illustrating the background, we analyze in particular a genome-scale reconstruction of the metabolism of the bacterium *E. coli* and illustrate the mechanisms in simple toy models.

## II. BACKGROUND

Here we consider here a simple model in which metabolism is regarded as a well-mixed chemical reaction network in the steady state and continuum limit. In the case of $M$ metabolites transformed through $N$ reactions, the network structure is encoded in a stoichiometric matrix $\mathbf{S} = \{S_{i\mu}\}$ whose entries are the stoichiometric coefficients of metabolite $\mu = 1 \dots M$ in reaction $i = 1 \dots N$. This matrix, through mass balance equations, enforces a linear relation between reaction fluxes $f_i$ and concentration $c_\mu$ variations of metabolites

$$\dot{c}_\mu = \sum_i S_{i\mu} f_i. \tag{1}$$

Once a steady state is reached, i.e., for time scales longer than diffusion and typical turnover times, and physical limits (reversibility, medium, kinetics) are included in terms of flux bounds, the reactions fluxes span a convex polytope:

$$\sum_i S_{i\mu} f_i = 0, \tag{2}$$

$$f_i \in [f_{i,\min}, f_{i,\max}]. \tag{3}$$

When flux bounds are not provided, it is customary to set them to an arbitrary large number, $f_i \in [-C, C]$ (typically $C = 10^3, 10^4$), that, for a meaningful model, will not influence the results. In the most simple way thermodynamics is implemented by setting reaction reversibility, i.e., $f_i \geqslant 0$ for some reactions. On the other hand a more rigorous yet simple approach consists in postulating that fluxes shall follow a free energy gradient [15,17,18]. If $f_i \neq 0$, then $f_i \Delta G_i < 0$ (Gibbs

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inequality) where $\Delta G_i$ is the free energy drop of reaction $i$. The reaction free energies can be written as the difference between the chemical potentials $g_\mu$ of products and substrates through the stoichiometric matrix $\Delta G_i = \sum_\mu S_{i\mu} g_\mu$. In terms of chemical potentials we have thus a system of linear inequalities $[\xi_{i\mu} = -sign(f_i) S_{i\mu}]$

$$\sum_\mu \xi_{i\mu} g_\mu > 0 \quad \forall i \tag{4}$$

whose feasibility (existence of a solution $g_\mu$) is necessary for the thermodynamical soundness of the flux configuration. From a mathematical viewpoint, the existence of solutions in systems of linear inequalities is ruled by duality theorems of the alternative, by which a necessary and sufficient condition for feasibility is the unfeasibility of a suitably defined dual linear system of inequalities. In particular for homogeneous strict inequalities the Gordan theorem holds, and the dual systems of (4) reads

$$\sum_i \xi_{i\mu} k_i = 0 \quad k_i \geqslant 0 \quad \forall \mu. \tag{5}$$

One and only one of the systems (4) or (5) will thus have nontrivial solutions. The solutions of the latter system have the form of closed cycles, whose existence alters in an artificial way flux scales. In fact, if $f_{i,0}$ is a feasible state and $k_i$ is a solution of (5), the line $f_i(L) = f_{i,0} + k_i s_i L$ verifies the steady state mass balance constraints by construction, and it will be inside the polytope till $L$ possibly reaches the arbitrary bounds; see Ref. [19] for further details. From a physical point of view they correspond to "perpetuum mobiles" from which it would be possible to get work for free (see Ref. [17] for an example with ATP production).

A simple illustration is provided in the toy network depicted in Fig. 1(a), where metabolite $A$ is injected in the system with rate $u$. Upon implementing mass balance constraints, $A$ is transformed either in $B$ with rate $x$ (subsequently transformed in $C$ with the same rate) or directly in $C$ with rate $u - x$, which is depleted with rate $u$. The flow can be inverted if $u < 0$ (i.e., with $C$ injected and $A$ depleted). The crucial point is that $x$ has no other bounds and reaches unfeasibly high values without taking into account properly thermodynamics, whose implementation leads on the other hand to the nonlinear constraints $ux \geqslant 0$, $|x| \leqslant |u|$ [17,18].

In essence thermodynamics forbids flux directions that gives solutions to system (5). This forbids a set of orthants within the feasible space that get decomposed in a connected star of convex polytopes that is overall not convex anymore [see Fig. 1(b)], but whose scales now shall reflect the truly physical constraints of the given model.

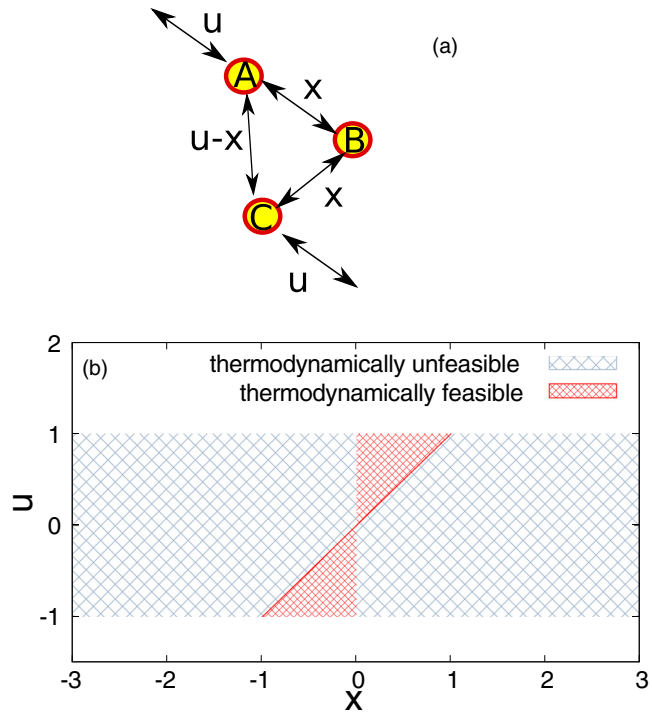


FIG. 1. Thermodynamic constraints give scale to stationary metabolic models. (a) Toy network model illustrating the mechanism; (b) the plane $(x,u)$, $u \in [-1,1]$, and $x \in [-3,3]$. Thermodynamics constraints impose further $ux \geqslant 0$ and $|x| \leqslant |u|$. Notice that the thermodynamically feasible space is not convex, it scales with the bounds of $u$ and leads to bimodal distributions.

## III. RESULTS

### A. *E. coli* genome scale model analysis

The example illustrated in the previous section shows how the implementation of thermodynamic constraints can set the scale in a stationary metabolic network model, a task otherwise hampered in presence of unfeasible loops, and furthermore it shows how this implementation leads to multimodal flux distributions. In large genome-scale metabolic network models we conjecture that the former observation could be simply at the origin of inferred scale-free tails in flux distributions. In this section these issues will be explored in a genome-scale metabolic network model. We consider here the typical steady states of the *E. coli* metabolic network iJR904 [20] in a glucose-limited minimal medium in aerobic conditions. The uniform sampling of the underlying convex polytope has been performed by a hit-and-run Monte Carlo Markov chain [21], whose convergence is guaranteed in feasible times [22] upon handling ill-conditioning [23]. Thermodynamically unfeasible cycles have been enumerated for this network [17], and they can be detected and removed in many ways [24–26]. They have been corrected with one of the methods exposed in Ref. [19]. Specifically, once an unfeasible loop is detected in a flux configuration, this is corrected removing the loop by adjusting the aforementioned parameter $L$ in order to silence the less intense reaction in the loop. As explained in Ref. [19] this correction is minimal in the sense that it corresponds to a local minimization of the $L_1$ norm and to a projection on the boundary of the thermodynamically feasible space. Results for the distribution of flux intensities $|f_i|$ are shown in Fig. 2(a), before and after correcting for thermodynamic unfeasibilities. The long tail corresponding to the uniform sampling depends on the arbitrary constant fixed for flux bounds ($C = 10^3$, $5 \times 10^3$, $10^4$ in the figure), and it can be extended thus arbitrarily. In contrast, after thermodynamic correction, the flux intensity distribution is independent from this arbitrary constant. In regard to single

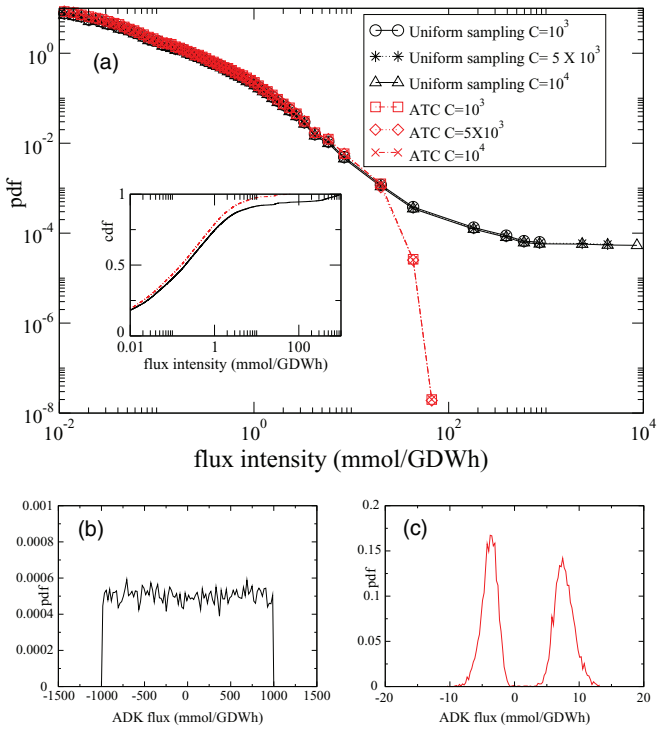


FIG. 2. (a) Overall flux intensity distribution in log-log scale before and after thermodynamic correction of typical steady states of the model of the metabolism of *E. coli* under study upon sampling $R = 10^5$ configurations ($N = 666$ reactions, dimension $D = 233$). The maximum limiting glucose uptake is $u = 6$ mmol/gDWh, while the arbitrary constant for reaction bounds is varied $C = 10^3$, $5 \times 10^3$, $10^4$. The flux intensity distribution depends on this constant before thermodynamic correction. ATC: after thermodynamic correction. Inset: cumulative distribution function, obtained by ranking, comparing the uniform sampling before and after thermodynamic correction ($C = 10^3$). (b, c) Histogram of adenylate kinase flux before (b) and after thermodynamic correction (c).

reaction flux distributions, we show in Fig. 2(b) the retrieved histogram of adenylate kinase, an important enzyme involved in energy homeostasis performing the reaction $2ADP \leftrightarrow ATP + AMP$. Since this reaction is involved in an unfeasible cycle [17], its distribution is flat and wide upon uniform sampling, while after thermodynamic correction shows a bimodal fashion opening possible speculations about the presence of two effective states for energetic homeostasis.

In order to keep track in a more rigorous way of the scales of the metabolic space in dependence on physical limiting factors, a principal component analysis of the sampled configuration has been performed upon varying glucose uptake conditions. The square root of the retrieved eigenvalues are proportional to the diameters of the maximum inscribed ellipsoid into the convex hull of the space and thus quantify its scales more rigorously.

We show in Fig. 3(a) the 100th largest eigenevalues before and after thermodynamic correction. At odds with the interesting case of assets cross-correlations in financial markets [27], here the largest eigenvalues for the uniform sampling correspond to unfeasible cycles that disappear after thermodynamic correction.

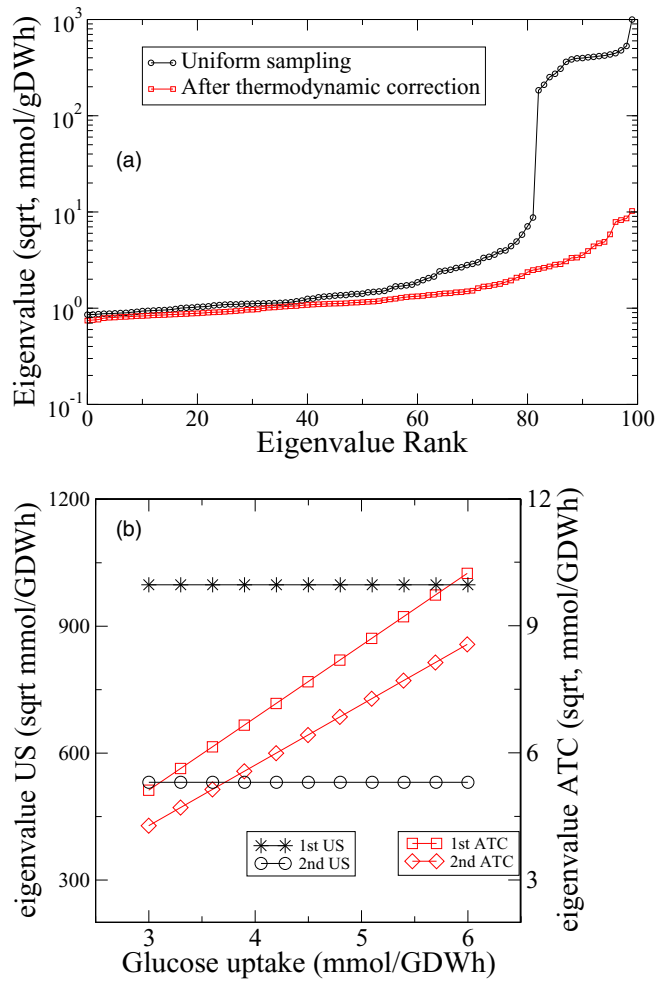


FIG. 3. (a) The 100th largest eigenvalues (square root) from a principal component analysis of the flux sampling before and after thermodynamics correction. (b) First and second largest eigenvalues (square root) as a function of the maximum glucose uptake from uniform sampling (US) and after thermodynamic correction (ATC, different $y$ scale).

In Fig. 3(b) it is shown that the two highest eigenvalues scale linearly with the maximum glucose uptake after taking into account thermodynamics, while they do not in the infeasible case, which is indeed scale free.

### B. Towards symmetry breaking in toy models

We have seen that the implementation of thermodynamic constraints makes in general the space of feasible steady states of chemical reaction networks nonconvex. This property would be interesting from a modeling perspective since one of the most widely employed techniques to model metabolic data, flux balance analysis, relies on the maximization of an objective function, typically a linear function of fluxes, that endows a biological significance as the growth rate and/or yield. In the simple case of a convex polytope [Eqs. (2) and (3)], this task computationally amounts to a well-known linear programming problem, and it gives simply interpretable solutions, i.e., typically one solution, that is a vertex of the polytope or at most a convex subset. On the other hand, for

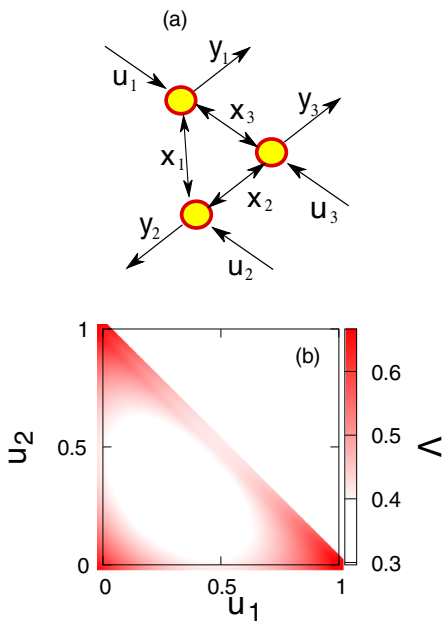


FIG. 4. Thermodynamic constraints and optimization potentially leading to symmetry breaking. (a) Toy network model illustrating the mechanism; (b) contour plot of the phase space volume $V$ in the plane $(u_1,u_2)$ when $U = 1$.

a nonconvex space, this optimization process could give rise to symmetry breaking, where multiple disconnected solutions arise. This mechanism could be relevant in modeling cell-to-cell interactions, as a means to explain and provide for effective heterogeneity. In this subsection this issue will be explored by considering the following simple toy model depicted in Fig. 4(a). In this model three units ("cells") intake, use, and/or exchange a given substrate. They uptake the substrate each with intensity $u_i$ and compete for it, i.e., there is a fixed constant $U$ that stands for limited resource availability $\sum_i u_i \leqslant U$. We will then indicate the cell usage rate $y_i$ and the exchange rate $x_i$ (to be considered from cell $i-1$ to cell $i$ if positive; and from $i$ to $i-1$ if negative).

We have the constraints (where we identify the index $4 = 1$)

$$u_i \geqslant 0, \quad \sum_i u_i \leqslant U, \quad u_i = y_i + x_{i+1} - x_i, \quad y_i \geqslant 0. \tag{6}$$

It should be noticed that such constraints alone do not bound the exchange rates, to which an arbitrary constant can be added $x_i \to x_i + R \quad \forall i$ without violations. On the other hand it is straightforward to see that if the $x_i$ would have the same sign, this would lead to a violation of thermodynamics (no free energy gradient) and further nonlinear constraints shall be added in turn to (6). Upon considering an optimization process in which cells maximize their output, i.e., we consider the linear objective $\sum_i y_i$, we see that $\sum_i y_i = \sum_i u_i \to U$. This leads to a degenerate solution space with each configuration $u_i$ having a phase space volume (to which probability is proportional in case of a flat prior)

$$V(\{u_i\}) = \int_{x_{i+1}-x_i \leqslant u_i} \prod dx_i \quad \text{(not all the same sign)}. \tag{7}$$

The integral can be evaluated for the simple case of three cells upon dividing it in the six integrals in which the sign $s_i$ of the $x_i$ is fixed (we exclude the two configurations with equal signs). For instance, consider the integral $I$ upon fixing $s_1 = +1$ and $s_2 = s_3 = -1$. We replace the $x_i$ with positive definite variables $r_i \geqslant 0$: $x_i = s_i r_i$. We have the constraints (from $x_{i+1} \leqslant x_i + u_i$)

$$-r_1 - r_2 \leqslant u_1, \tag{8}$$

$$r_2 - r_3 \leqslant u_2, \tag{9}$$

$$r_3 + r_1 \leqslant u_3. \tag{10}$$

The first constraint is trivially satisfied, and we can integrate $0 \leqslant r_1 \leqslant u_3 - r_3$ and $0 \leqslant r_2 \leqslant u_2 + r_3$, obtaining

$$I = \int_0^{u_3} (u_3 - r_3)(u_2 + r_3)\, dr_3 = \frac{u_3^2}{2}(u_2 + u_3/3). \tag{11}$$

Considering the other cases that can be obtained by symmetry we have finally

$$2V(u_1,u_2,u_3) = U \sum_i u_i^2 - \frac{1}{3} \sum_i u_i^3. \tag{12}$$

This function has a minimum for the symmetric solution $u_i = U/3$ and three (equal) maxima where one cell is uptaking all resources, e.g., $u_1 = U$ and $u_2 = u_3 = 0$, that turns out to be the most probable outcome [Fig. 4(b)].

## IV. CONCLUSIONS

Although scientists have established it to be impossible under the laws of physics, perpetual motion continues to capture the imagination of inventors and to enter in a subtle way in scientific research as well. In this work it has been shown that in order to meaningfully assess scales in metabolic networks it would be necessary to take into account thermodynamics, given the free energy-transducing nature of these systems. It has been shown in one realistic example that the long tails inferred in flux distributions in metabolic network models are a spurious effect due to reactions involved in infeasible cycles. From the positive side, such an energy balance analysis can be rewarding from several aspects, since, apart from assessing metabolites' concentrations [28], free energies [29], and potentially regulated sites [30], it reveals that the metabolic space has a more complex and richer [31] multimodal structure. The additional layer of complexity provided by thermodynamics could enrich the possibility of biophysical modeling especially in presence of heterogeneity, cell-to-cell interactions [32], and/or metabolic shuttling and could potentially lead to new theoretical insights [33,34]. A simple toy model has been provided to show a possible mechanism of symmetry breaking while maximizing for an objective function. Toy models have been used to illustrate in a tractable setting how the implementation of thermodynamic constraints relates to scales, nonconvexity, and symmetry breaking, a picture that has been confirmed on a genome-scale model, but further work is certainly needed to set up in a more rigorous way these issues.

Finally it should be remarked that the inference from such constraints is challenging from a computational point of view given the nonconvexity of the space [35]. Suitable extensions of fast approximate analytical methods for the convex case [36,37] could be very promising in this respect.

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[1] G. Galilei, *Galileo: Two New Sciences (Translation by Stillman Drake of Galileo's 1638 Discourses and Mathematical Demonstrations Concerning Two New Sciences)* (University of Wisconsin Press, Madison, 1974).

[2] J. Cardy, *Scaling and Renormalization in Statistical Physics*, Vol. 5 (Cambridge University Press, Cambridge, 1996).

[3] E. Schneidman, M. J. Berry, R. Segev, and W. Bialek, Weak pairwise correlations imply strongly correlated network states in a neural population, Nature (London) **440**, 1007 (2006).

[4] A. Levina, J. M. Herrmann, and T. Geisel, Dynamical synapses causing self-organized criticality in neural networks, Nat. Phys. **3**, 857 (2007).

[5] I. Mastromatteo and M. Marsili, On the criticality of inferred models, J. Stat. Mech.: Theory Exp. (2011) P10012.

[6] G. Tkačik, T. Mora, O. Marre, D. Amodei, S. E. Palmer, M. J. Berry, and W. Bialek, Thermodynamics and signatures of criticality in a network of neurons, Proc. Natl. Acad. Sci. USA **112**, 11508 (2015).

[7] N. Lane, *The Vital Question: Energy, Evolution, and the Origins of Complex Life* (W. W. Norton, New York, 2015).

[8] J. D. Orth, I. Thiele, and B. Ø Palsson, What is flux balance analysis? Nat. Biotechnol. **28**, 245 (2010).

[9] K. R. Patil, M. Åkesson, and J. Nielsen, Use of genome-scale microbial models for metabolic engineering, Current Opin. Biotech. **15**, 64 (2004).

[10] E. Almaas, B. Kovacs, T. Vicsek, Z. N. Oltvai, and A.-L. Barabási, Global organization of metabolic fluxes in the bacterium Escherichia coli, Nature (London) **427**, 839 (2004).

[11] J. Monod, The growth of bacterial cultures, Annu. Rev. Microbiol. **3**, 371 (1949).

[12] H. Bremer and P. P. Dennis, Modulation of chemical composition and other parameters of the cell at different exponential growth rates, EcoSal Plus **3** (2008).

[13] D. A. Beard, S.-d. Liang, and H. Qian, Energy balance for analysis of complex metabolic networks, Biophys. J. **83**, 79 (2002).

[14] M. Ataman and V. Hatzimanikatis, Heading in the right direction: Thermodynamics-based network analysis and pathway engineering, Current Opin. Biotech. **36**, 176 (2015).

[15] A. C. Reimers, Metabolic networks, thermodynamic constraints, and matroid theory, Ph.D. thesis, Freie Universität Berlin, 2014.

[16] M. Polettini and M. Esposito, Irreversible thermodynamics of open chemical networks. I. Emergent cycles and broken conservation laws, J. Chem. Phys. **141**, 024117 (2014).

[17] D. De Martino, Thermodynamics of biochemical networks and duality theorems, Phys. Rev. E **87**, 052108 (2013).

[18] E. Noor, N. E. Lewis, and R. Milo, A proof for loop-law constraints in stoichiometric metabolic networks, BMC Syst. Biol. **6**, 140 (2012).

[19] D. D. Martino, F. Capuani, M. Mori, A. D. Martino, and E. Marinari, Counting and correcting thermodynamically infeasible flux cycles in genome-scale metabolic networks, Metabolites **3**, 946 (2013).

[20] J. L. Reed, T. D. Vo, C. H. Schilling, and B. O. Palsson, An expanded genome-scale model of *Escherichia coli* k-12 (iJR904 GSM/GPR), Genome Biol. **4**, R54 (2003).

[21] V. Turcin, On the computation of multidimensional integrals by the Monte Carlo method, Theory Probab. Appl. **16**, 720 (1971).

[22] L. Lovász, Hit-and-run mixes fast, Math. Program. **86**, 443 (1999).

[23] D. D. Martino, M. Mori, and V. Parisi, Uniform sampling of steady states in metabolic networks: Heterogeneous scales and rounding, PLoS ONE **10**, e0122670 (2015).

[24] J. Schellenberger, N. E. Lewis, and B. Ø. Palsson, Elimination of thermodynamically infeasible loops in steady-state metabolic models, Biophys. J. **100**, 544 (2011).

[25] D. D. Martino, M. Figliuzzi, A. D. Martino, and E. Marinari, A scalable algorithm to explore the Gibbs energy landscape of genome-scale metabolic networks, PLoS Comput. Biol. **8**, e1002562 (2012).

[26] P. A. Saa and L. K. Nielsen, ll-achrb: A scalable algorithm for sampling the feasible solution space of metabolic networks, Bioinformatics **32**, 2330 (2016).

[27] M. Marsili, Dissecting financial markets: Sectors and states, Quant. Finance **2**, 297 (2002).

[28] A. Hoppe, S. Hoffmann, and H. Holzhutter, Including metabolite concentrations into flux balance analysis: Thermodynamic realizability as a constraint on flux distributions in metabolic networks, BMC Syst. Biol. **1**, 23 (2007).

[29] C. Henry, L. Broadbelt, and V. Hatzimanikatis, Thermodynamics-based metabolic flux analysis, Biophys. J. **92**, 1792 (2007).

[30] A. Kümmel, S. Panke, and M. Heinemann, Putative regulatory sites unraveled by network-embedded thermodynamic analysis of metabolome data, Mol. Syst. Biol. **2**, 2006.0034 (2006).

[31] R. Tanaka, Scale-Rich Metabolic Networks, Phys. Rev. Lett. **94**, 168101 (2005).

[32] M. Di Filippo, C. Damiani, R. Colombo, D. Pescini, and G. Mauri, Constraint-based modeling and simulation of cell populations, arXiv:1607.04463.

[33] D. D. Martino, F. Capuani, and A. D. Martino, Inferring metabolic phenotypes from the exometabolome through a thermodynamic variational principle, New J. Phys. **16**, 115018 (2014).

[34] Ö. Kartal, S. Mahlow, A. Skupin, and O. Ebenhöh, Carbohydrate-active enzymes exemplify entropic principles in metabolism, Mol. Syst. Biol. **7**, 542 (2011).

[35] A. C. Reimers, Obstructions to sampling qualitative properties, PloS ONE **10**, e0135636 (2015).

[36] J. Fernandez-de-Cossio-Diaz and R. Mulet, Fast inference of ill-posed problems within a convex space, J. Stat. Mech.: Theory Exp. (2016) 073207.

[37] A. Braunstein, A. P. Muntoni, and A. Pagnani, An analytic approximation of the feasible space of metabolic networks, Nat. Commun. **8** (2017).